EMS Communication Protocol

(Modbus TCP)

V1.7.1

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Modification History

Version	Date of release	Modifier	Main changes
1.0	2024/05/31	Song Junjie	First version
1.1	2024/09/28	Li Qi	1. Add hardware version information and SN number 2. Update the ems meter information (place important real-time power information and status information in the overview information. 3. Add device status Settings and power Settings 4. Add heart rate data (not supported yet) 5. Add handshake and encryption (not supported yet)
1.2	2024/10/28	Li Qi	Add bms and pcs alerts
1.3	2024/12/27	Xiang Xiaolong	Modify the register description according to the actual PCS
1.4	2025/03/03	Li Qi	Add bms and pcs data
1.5	2025/04/07	Song Junjie	Add Charge/Discharge StopSoc Limit R/W Registers, BMS/PCS Trouble Code, and English Translation
1.6	2025/04/14	Li Qi	Add electricity meter data
1.7	2025/04/28	Xiang Xiaolong	Add single-phase active power control, reactive power control
1.7.1	2025/9/5	Zilong Zhu	**Rectify the mistakes within the amended document**

Catalogue

1. Summary of communications

1.1 Scope of application of the Agreement

This agreement is used for the EMS of the ESS system to communicate through Ethernet TCP/IP.

2. Protocol Specification

2.1 Protocol reference

GB/Z 19582.1-2004 Industrial Automation Network Specification Based on Modbus Protocol Part 3.

The protocol is generally used for TCP/IP, communication link, master-slave sites, and communication protocols that comply with the modbus specification;

2.2 Communication methods

Ethernet communication, TCP/IP protocol, EMS as server mode, for access by upper machine in client mode or other software systems;

2.3 IP addresses

EMS Address：

eth0: 192.168.1.24 （ Back-up）

eth1: 192.168.1.15 （ Local ethnet）

eth2: 192.168.3.15 （ Back-up）

eth3: (DHCP internet）

2.4 Port name

EMS as server mode, port number: 502;

2.5 Data format agreement

For multi-byte data sending and receiving, the big-end mode (Big Endian) is used, that is, the high byte is sent first and the low byte is sent later.

2.6 Alarm code level

Level I > Level II > Level III

3. Regulation Description

3.1 Modbus frame format

The protocol for a Modbus serial line is a master-slave protocol where at any given time only one master node is connected to the bus and one or more (up to 247) slave nodes are also connected to the same serial bus. A Modbus communication always starts with the master node and no slave nodes are received.

MBPA masthead				PDU	
Transaction identifier	**Protocol identifier**	**Data Length**	**Unit identifier /address**	**Code**	**Data**
2 bytes	2 bytes	2 bytes	1 byte	1 byte	N bytes

3.2 Functional codes

No.	Code	Description	Note
1	0x03	Read multiple holding registers	
2	0x06	Write a single holding register	
3	0x10	Write multiple holding registers	

4. Modbus Response Description

4.1 Function code 0x03

Used to read a series of registers that are kept in memory.

Server request format：

Transaction identifier		Protocol identifier		Data Length	identifier/address	Code	Start address	Number of registers
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x03	2 Bytes	2 Bytes

The client responds normally:

Transaction identifier		Protocol identifier		Data Length	identifier/address	Code	Start address	Number of registers
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x03	1 Byte	N Bytes

The client responds abnormally：

Transaction identifier		Protocol identifier		Data Length	identifier/address	Request function code	Error code
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x83	1 Byte

Note: Data length = number of registers * 2.

4.2 Function code 0x06

Write a single holding register.

Server request format：

Transaction identifier		Protocol identifier		Data Length	identifier/address	Code	Start address	Number of registers
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x06	2 Bytes	2 Bytes

The client responds normally:

The client responds normally: Transaction identifier		Protocol identifier		Data Length	identifier/address	Code	Start address	Number of registers
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x06	2 Bytes	2 Bytes

The client responds abnormally：

Transaction identifier		Protocol identifier		Data Length	identifier/address	Request function code	Error code
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x86	1 Byte

4.3 Function code 0x10

Write multiple continuous holding registers.

Server request format：

Transaction identifier		Protocol identifier		Data Length	identifier/ address	Code	Start address	Number of registers	DL	Data
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x10	2 Bytes	2 Bytes	1 Byte	N Bytes

Client normal response (EMS reply)：

Transaction identifier		Protocol identifier		Data Length	identifier/address	Code	Start address	Number of registers
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x10	2 Bytes	2 Bytes

Abnormal response from the client

Transaction identifier		Protocol identifier		Data Length	identifier/ address	Request function code	Error code
0x00	0x00	0x00	0x00	2 Bytes	0x01	0x90	1 Byte

4.4 Error Code

Code_1 0x01 Function code is invalid
Code_2 0x02 Request address is invalid
Code_3 0x03 Register count is invalid
Code_4 0x04 Register data is invalid
Note: The above "starting address" and "ending address" are inclusive, and the address is a decimal number.

5. EMS Register table information

5.1 Version information(Read only)

Address	Type	Description	Value	Note
0001	UINT16	Product type	1	1-EMS
0002	UINT16	(BaseMajor) Software version identifier		value
0003	UINT16	(BaseMinor) Software version identifier		value
0004	UINT16	(ProjetMajor) Software version identifier		value
0005	UINT16	(ProjetMinor) Software version identifier		value
0006	UINT16	(BaseMajor) Hardware version identifier		value
0007	UINT16	(BaseMinor) Hardware version identifier		value
0008	UINT16	SN_1	“EA”	ASCII
0009	UINT16	SN_2	“1H”	ASCII
0010	UINT16	SN_3		ASCII
0011	UINT16	SN_4		ASCII
0012	UINT16	SN_5		ASCII
0013	UINT16	SN_6		ASCII
0014	UINT16	SN_7		ASCII

Note: The above data only supports the 03 function code reading.

5.1.1 Software and hardware version numbers

Software version number, Example:1.0.2.0

BaseMajor: 1

BaseMinor: 0

ProjetMajor: 2

ProjetMinor: 0

5.1.2 SN number

e.g.: “EA1H240108001”

The corresponding ASCII HEX representation is:
45 41 | 31 48 | 32 34 | 30 31 | 30 38 | 30 30 | 31 00
SN_1: 0x4541 SN_2: 0x3138 SN_3: 0x3234 SN_4: 0x3031
SN_5: 0x3038 SN_6: 0x3030 SN_7: 0x3100

5.2 Overview data（Read Only）

Address	Type	Description	Unit	Note
1001	UINT16	EMS Running status	/	0- not ready 1- Stop(Shut down or inactive/disabled) 2- Start in progress 3- Normal (control mode = 2, allow to be controlled) 4- Fault, unable to run
1002	UINT16	Power control type	/	0- Invalid 1- Battery（Ctrl BMS power） 2- Ctrl Grid emeter power)
1003	UINT16	control mode	/	0- Invalid 1- local manual 2- remote control
1004	UINT16	Total ESS-SOC	0.1%	0~1000
1005	UINT16	Total ESS-current maximum allowable charging power	0.1kW	Current real-time maximum charging capacity
1006	UINT16	Total ESS-current maximum allowable discharge power	0.1kW	Current real-time maximum discharge capacity
1007	INT16	ESS real-time total power	0.1kW	charge:+; discharge:-
1008	UINT16	Grid meter-operation status	/	0- Invalid 1- offline 2- online
1009	INT16	Grid meter-real-time active power	0.1kW	+:forward -:reverse (forward: power from grid, reverse:power to grid)
1010	UINT16	Total load-operation status	/	0- Invalid 1- offline 2- online
1011	INT16	Total load-real-time active power	0.1kW	+:forward -:reverse (including all loads)
1012	UINT16	Total PV-operation status	/	0- Invalid 1- Offline 2- Stop 3- Alam 4- Fault 5- Running
1013	INT16	Total PV-total output active power	0.1kW	Positive or 0
1014	UINT16	Reserved	/	

Note: The above data only supports 03 function code reading.

Power parameter agreement: power goes into the system is positive(+), out is negative（-）.

5.3 EMS Detailed Data (Read Only)

Address	Type	Description	Unit	Note
1101	INT16	Total ESS real-time total reactive power	0.1kVar	
1102	UINT16	Total ESS-charged energy amount in current day	0.1kWh	
1103	UINT16	Total ESS-discharged energy amount in current day	0.1kWh	
1104	INT16	Grid meter-real-time total reactive power	0.1kVar	
1105	UINT16	Grid meter-charged energy amount in current day	0.1kWh	
1106	UINT16	Grid meter-discharged energy amount in current day	0.1kWh	
1107	INT16	Total PV-total output reactive power	0.1kVar	
1108	UINT16	Total PV-energy generated in current day	0.1kWh	
1109	UINT16	Total Load cost energy in current day	0.1kWh	
1110	UINT16	Reserved	/	
1201	UINT32	Total ESS-rated capacity	0.1kWh	
1203	UINT32	Total ESS-cumulative charged energy amount	0.1kWh	
1205	UINT32	Total ESS-cumulative discharged energy amount	0.1kWh	
1207	UINT32	Grid meter-cumulative positive energy amount	0.1kWh	
1209	UINT32	Grid meter-cumulative reverse energy amount	0.1kWh	
1211	UINT32	Total load-cumulative positive energy amount	0.1kWh	
1213	UINT32	Total load-cumulative reverse energy amount	0.1kWh	
1215	UINT32	Reserved	/	
1401	UINT16	BMS Fault	Bits	0-Normal 1-Fault bit0: BMS_1 bit1: BMS_2 bit2: BMS_3 bit3: BMS_4 bit4: BMS_5 …… Reserved

1402	UINT16	PCS Fault	Bits	0-Normal 1-Fault bit0: PCS_1 bit1: PCS_2 bit2: PCS_3 bit3: PCS_4 bit4: PCS_5 …… Reserved
1403	UINT16	EMeter Fault	Bits	0-Normal 1-Fault bit0: Grid Emeter bit1: Ess Emeter bit2: PV Emeter bit3: Load Main Emeter bit4: Load Sub Emeter bit5: Other Emeter …… Reserved
1404	UINT16	PV Solar Fault	Bits	0-Normal 1-Fault bit0: PV_1 bit1: PV_2 bit2: PV_3 bit3: PV_4 bit4: PV_5 …… Reserved
1405	UINT16	DCDC Fault	Bits	0-正常Normal 1-故障Fault bit0: DCDC_1 bit1: DCDC_2 bit2: DCDC_3 bit3: DCDC_4 bit4: DCDC_5 …… Reserved
1406	UINT16	Reserved	/	/

Note：The above data only supports the 03 function code for reading.

6. Device information

6.1 BMS

Address: 40000-50000 (Decimal system (Dec))
Offset address
P = 40000 + 1000*n
(n represents the number of ABMS/CBMS, 0≤n<10, n=0 represents the first ABMS/CBMS)

Each battery stack has 200 register addresses, with a maximum total of 10 supported

Address	Type	Description	Unit	Note
P+0	UINT16	Device Manufacturer	/	
P+1	UINT16	Device Enable Flag	/	0- Disable 1- Enable
P+2	UINT16	Communication Status Flag	/	0- Invalid 1- Error 2- Offline 3- Online
P+3	UINT16	Communication Heartbeat Signal	/	
P+4	UINT16	Battery Status	/	0- Not Ready 1- Initalization 2- Starting 3- Running
P+5	UINT16	Fault Flag	/	0- Invalid 1- Idle 2- Warn 3- Fault
P+6	UINT16	Battery Charge/Discharge Status	/	0- Idle 1- Charging 2- DisCharging
P+7	UINT16	Stack Circuit Breaker Status	/	0- Invalid 1- Open 2- Close
P+8	UINT16	Cluster Circuit Breaker Status	/	0- Invalid 1- Open 2- Close
P+9	UINT16	Cluster Relay Closure Status	/	0- Invalid 1- Open 2- Close
P+10	UINT16	Accumulated Total Voltage	0.1V	
P+11	UINT16	Measured Total Voltage	0.1V	
P+12	INT16	Total Current	0.1A	
P+13	INT16	Total Power	0.1kW	
P+14	UINT16	SOC State of Charge (SOC)	0.1%	
P+15	UINT16	SOH State of Health (SOH)	0.1%	

P+16	UINT16	Average Cell Voltage	mV	
P+17	INT16	Average Battery Temperature	℃	
P+18	INT16	Maximum Charging Power Limit	0.1kW	
P+19	INT16	Maximum Discharging Power Limit	0.1kW	
P+20	UINT16	Remaining Chargeable Energy	0.1kWh	
P+21	UINT16	Remaining Dischargeable Energy	0.1kWh	
P+22	UINT16	Highest Cell Cluster Number (starting from 1)	/	
P+23	UINT16	Highest Cell Pack Number (starting from 1)	/	
P+24	UINT16	Highest Cell ID (valid starting from 1)	/	
P+25	UINT16	Maximum Cell Voltage Value	mV	
P+26	UINT16	Lowest Cell Cluster Number (starting from 1)	/	
P+27	UINT16	Lowest Cell Pack Number (starting from 1)	/	
P+28	UINT16	Lowest Cell ID (valid starting from 1)	/	
P+29	UINT16	Minimum Cell Voltage Value	mV	
P+30	UINT16	Maximum Voltage Difference	mV	
P+31	UINT16	Highest Cell Temperature Cluster Number (starting from 1)	/	
P+32	UINT16	Highest Cell Temperature Pack Number (starting from 1)	/	
P+33	UINT16	Highest Battery Temperature ID (valid starting from 1)	/	
P+34	UINT16	Maximum Battery Temperature Value	℃	
P+35	UINT16	Lowest Cell Temperature Cluster Number (starting from 1)	/	
P+36	UINT16	Lowest Cell Temperature Pack Number (starting from 1)	/	
P+37	UINT16	Lowest Battery Temperature ID (valid starting from 1)	/	
P+38	UINT16	Minimum Battery Temperature Value	℃	
P+39	UINT16	Maximum Temperature Difference	℃	
P+40	UINT16	Daily Charged Energy	0.1kWh	
P+41	UINT16	Daily Discharged Energy	0.1kWh	
P+42	UINT32	Accumulated Charged Energy	0.1kWh	

P+44	UINT32	Accumulated Discharged Energy	0.1kWh	
P+46	UINT32	Reserved		
P+48	UINT32	Reserved		
P+50	UINT32	BMS Alarm Level I	Bits	BMSAlarmLevel_I_II_III
P+52	UINT32	BMS Alarm Level II	Bits	BMSAlarmLevel_I_II_III
P+54	UINT32	BMS Alarm Level III	Bits	BMSAlarmLevel_I_II_III
P+56	UINT32	CBMS Fault	Bits	CbmsFault
P+58	UINT32	ABMS Fault	Bits	AbmsFault
P+60	UINT32	BMS Peripheral Fault	Bits	BMSPeripheralFault
P+62	UINT32	Air Cooll Alarm	Bits	AirCoolAlarm
P+64	UINT32	Liquid Cool Alarm	Bits	LiquidCoolAlarm
P+66	UINT32	Fire Fighting Alarm	Bits	FirefightingAlarm
P+（180-199）	UINT16	Battery Pack Voltage (Total Voltage of Modules 1-20)	V	20
P+（200-599）	UINT16	Cell Voltages per Module (20 Modules, 20 Cells per Module)	V	20*20=400
P+（600-999）	INT16	Temperatures per Module (20 Modules, 20 Cells per Module)	℃	20*20=400

6.2 PCS

Address： 15000-15999（DEC）
Offset address:
P = 15000 + 100*n
(n represents the number of PCS, 0≤n<10, n=0 represents the first PCS)

Each PCS has 100 register addresses, with a maximum total of 10 supported

Address	Type	Description	Unit	Note
P+0	UINT16	Equipment Manufacturer	/	
P+1	UINT16	Device Enable Flag	/	0- Disable 1- Enable
P+2	UINT16	Communication Status Flag	/	0- Invalid 1- Communication Anomaly 2- Offline 3- Online
P+3	UINT16	Communication Module Thread Alive Signal	/	
P+4	UINT16	System Status	/	0- Not Ready 1- Inited,Ready for External Boot 2- Booting After Startup Command Sent 3- Running
P+5	UINT16	Fault Flag	/	0- Invalid 1- Normal 2- Alarm 3- Fault
P+6	UINT16	Device Power State	/	0- Invalid 0- Power On 1- Power Off
P+7	UINT16	On-grid/Off-grid Status	/	0- Invalid 1- PQ(Power Control) 2- VF(Voltage and Frequency) 3- VSG(Virtual Synchronous Generator)
P+8	UINT16	Reserve	/	
P+9	UINT16	Remote/Local Control Mode	/	0- Invalid 1- Local 2- Remote
P+10	INT16	Total Active Power	0.1kW	
P+11	INT16	Total Reactive Power	0.1kVar	
P+12	INT16	Total Apparent Power	0.1kVA	
P+13	UINT16	Total Power Factor	0.01	
P+14	INT16	Total Current	0.1A	

P+15	INT16	Grid Line Voltage	0.1V	
P+16	UINT16	Frequency	0.01Hz	
P+17	UINT32	Alarm Code - HW Hardware	Bits	AlarmHW
P+19	UINT32	Fault Code - HW Hardware	Bits	FaultHW
P+21	UINT32	Fault Code - AC Alternating Current	Bits	FaultAC
P+23	UINT32	Fault Code - DC Direct Current	Bits	FaultDC
P+25	UINT32	Fault Code - Sys System	Bits	FaultSYS
P+27	UINT32	Daily Charged Energy	0.1kWh	
P+29	UINT32	Daily Discharged Energy	0.1kWh	
P+31	UINT32	Accumulated Charged Energy	0.1kWh	
P+33	UINT32	Accumulated Discharged Energy	0.1kWh	
P+34	INT16	Active Power Phase A	0.1kW	
P+35	INT16	Reactive Power Phase A	0.1kVar	
P+36	INT16	Apparent Power Phase A	0.1kVA	
P+37	UINT16	Power Factor Phase A	0.01	
P+38	INT16	Current Phase A	0.1A	
P+39	INT16	Line Voltage A	0.1V	
P+40	INT16	Active Power Phase B	0.1kW	
P+41	INT16	Reactive Power Phase B	0.1kVar	
P+42	INT16	Apparent Power Phase B	0.1kVA	
P+43	UINT16	Power Factor Phase B	0.01	
P+44	INT16	Current Phase Phase B	0.1A	
P+45	INT16	Line Voltage B	0.1V	
P+46	INT16	Active Power Phase C	0.1kW	
P+47	INT16	Reactive Power Phase C	0.1kVar	
P+48	INT16	Apparent Power Phase C	0.1kVA	
P+49	UINT16	Power Factor Phase C	0.01	
P+50	INT16	Current Phase C	0.1A	

P+51	INT16	Line Voltage C	0.1V	
P+52	INT16	IGBT Temperature	℃	
P+53	INT16	Cabinet Temperature	℃	

6.3 Electricity Meter

Address： 16000-16999 (decimalism)

Offset Address:

P = 16000 + 100*n

(n represents the number of electricity meter, 0≤n<10, n=0 represents the first electricity meter)

Each PCS has 100 register addresses, with a maximum total of 10 supported

Register Address	Data Type	Data Content	Unit	Note
P+0	UINT16	Equipment Manufacturer	/	
P+1	UINT16	Device Enable Flag	/	0- Disable 1- Enable
P+2	UINT16	Communication Status Flag	/	0- Invalid 1- Communication Anomaly 2- Offline 3- Online
P+3	UINT16	Communication Module Thread Alive Signal	/	
P+4	UINT16	System Status	/	0- Idle 1- Initialization 2- Starting 3- Running
P+5	UINT16	Fault Flag	/	0- Invalid 1- Normal 2- Alarm 3- Fault
P+6	UINT16	Meter type		0- Invalid 1- Grid-side meter, gateway meter4 2- Load master meter 3- Load sub-meter 4- Energy storage meter 5- PV meter
P+7	INT16	Total Active Power	0.1kW	
P+8	INT16	Total Reactive Power	0.1kVar	
P+9	INT16	Total Apparent Power	0.1kVA	
P+10	UINT16	Total Power Factor	0.01	
P+11	INT16	Total Current	0.1A	
P+12	INT16	Grid Line Voltage	0.1V	
P+13	UINT16	Frequency	0.01Hz	
P+14	UINT16	Daily Charged Energy	0.1kWh	
P+15	UINT16	Daily Discharged Energy	0.1kWh	
P+16	UINT32	Accumulated Charged Energy	0.1kWh	

P+18	UINT32	Accumulated Discharged Energy	0.1kWh	
P+20	INT16	Active Power Phase A	0.1kW	
P+21	INT16	Reactive Power Phase A	0.1kVar	
P+22	INT16	Apparent Power Phase A	0.1kVA	
P+23	UINT16	Power Factor Phase A	0.01	
P+24	INT16	Current Phase A	0.1A	
P+25	INT16	Line Voltage A	0.1V	
P+26	INT16	Active Power Phase B	0.1kW	
P+27	INT16	Reactive Power Phase B	0.1kVar	
P+28	INT16	Apparent Power Phase B	0.1kVA	
P+29	UINT16	Power Factor Phase B	0.01	
P+30	INT16	Current Phase Phase B	0.1A	
P+31	INT16	Line Voltage B	0.1V	
P+32	INT16	Active Power Phase C	0.1kW	
P+33	INT16	Reactive Power Phase C	0.1kVar	
P+34	INT16	Apparent Power Phase C	0.1kVA	
P+35	UINT16	Power Factor Phase C	0.01	
P+36	INT16	Current Phase C	0.1A	
P+37	INT16	Line Voltage C	0.1V	
P+38	INT16	Rated limited power	0.1kW	
P+39	UINT16	Forward power limit	0.1kW	**Under normal circumstances,** = Rated limited power
P+40	UINT16	Reverse power limit	0.1kW	**Under normal circumstances,** = Rated limited power
P+41	UINT32	Alarm Code Level 1	Bits	not have
P+43	UINT32	Alarm Code Level 2	Bits	not have

7. Parameter and Status Settings (Read/Write)

Address	Type	Description	Range	Unit	Note
2001	UINT16	Enable External Scheduling	0.Disable 1.Enable	/	0-Default
2002	UINT16	ESS Charging SOC Upper Limit	0~1000	0.1%	95%-Default
2003	UINT16	ESS Discharging SOC Lower Limit	0~1000	0.1%	5%-Default

Note：

1.The above data only supports 10、 06 function codes

2.To schedule the local controller, you need to set register 2001 to 1 first (and maintain heartbeat activity, it will automatically reset to zero upon timeout)

3.To enable the SOC periodic calibration function, when powering on the user EMS, you need to modify R2002 and R2003 to the real interval corresponding to the EMS scheduled SOC

8. Real Time Power Control (Read/Write)

Address	Type	Description	Unit	Note
2501	UINT16	**Power Contrl Type**	/	1. Battery（ **Control PCS power）** 2. Grid(**Control Grid Power)**
2502	INT16	**Total Active Power value(+:charge,-:discharge)**	kW	**Note: When switching power types, the power automatically becomes 0, and it is necessary to reissue the power**
2503	INT16	**Toatal Reactive Power value(+:inductive,-:capacitive)**	kVar	

Note：: The above data supports the 06 and 10 function codes. The phase-by-phase power control at address 2502-2503 is only applicable to certain models; if this function is required, please contact POEROAD in advance.

9. Heart Beat

When EMS detects no read/write frames for over 30 seconds, it will terminate external task execution，and come back to local automatic mode.

10. Trouble Code Description（Read Only）

10.1. BMS Alarm Code

10.1.1. BMS Alarm Level I/II/III

Bit Position	Register Name
bit0	cell voltage high
bit1	cell voltage low
bit2	cell voltage diff high
bit3	total voltage high
bit4	total voltage low
bit5	cell temperature rise fast
bit6	charge temperature high
bit7	charge temperature low
bit8	cell temperature difference high
bit9	discharge temperature high
bit10	discharge temperature low
bit11	/
bit12	charge current high
bit13	discharge current high
bit14	insulation low
bit15	pack voltage high
bit16	pack voltage low
bit17	SOC high
bit18	SOC low
bit19	SOE high
bit20	SOE low
bit21	cluster voltage difference high
bit22	cluster current difference high
bit23	cluster charge current high
bit24	cluster discharge current high
bit25	cluster circulation high
bit26~31	/

10.1.2. CBMS Fault

Bit Position	Register Name
bit0	cell voltage fault
bit1	tempsensor open circuit
bit2	tempsensor short circuit
bit3	balance error
bit4	hardware error
bit5	cell voltage static error
bit6	cell voltage too high
bit7	Temperature too high
bit8	cell voltage low
bit9	cell voltage jump
bit10	cell temperature static permanent fault
bit11	cell temperature jump
bit12	/
bit13	/
bit14	/
bit15	/
bit16	cluster total voltage abnormal
bit17	HVM communication abnormal
bit18	ABMS communication fault
bit19	pack communication fault
bit20	circuit breaker fault
bit21	main positive relay fault
bit22	main negative relay fault
bit23	current abnormal
bit24	current limit abnormal
bit25	fuses_fault
bit26	short_circuit_current
bit27	fan fault
bit28	/
bit29	/
bit30	/
bit31	internal abnormal

10.1.3. ABMS Fault

Bit Position	Register Name
bit0	cluster communication abnormal
bit1	ABMS fan fault
bit2	ABMS breaker fault
bit3	Insufficient number of clusters ready
bit4	cluster number exceed
bit5	ABMS current loop fault
bit6	total volt Invalid
bit7	Temperature Invalid
bit8	ABMS_HVM communication abnormal
bit9	current limit fault
bit10	24V power volt abnormal
bit11	BAU hardware abnormal
bit12~30	/
bit31	internal_fault_all

10.1.4. BMS Peripheral Fault

Bit Position	Register Name
bit0	Emergency Power Off(EPO)
bit1	pcs_communication_failed
bit2	ems_communication_failed
bit3	fire_system_anomaly
bit4	liquid_cooling_unit_anomaly
bit5	air_conditioner_anomaly
bit6	water_leakage_anomaly
bit7	smoke_detector_anomaly
bit8	combustible_gas_anomaly
bit9	ups_anomaly
bit10	fire_alarm_warning
bit11	fire_alarm_activation
bit12	dc_lightning_protection_fault
bit13	ac_lightning_protection_fault
bit14	ups_discharge_complete
bit15	smoke_detector_alarm
bit16	co_concentration_high
bit17	h2_concentration_high
bit18	access_control_alert
bit19	smoke_detector_warning
bit20	water_leakage_warning
bit21	fire_system_alert
bit22	lightning_strike_alert
bit23	/
bit24	/
bit25	grid_power_outage_occur
bit26~30	/

10.1.5. Air Cool Alarm

Bit Position	Register Name
bit0	high_temp_alarm
bit1	low_temp_alarm
bit2	/
bit3	/
bit4	coil_anti_freeze
bit5	discharge_high_temp
bit6	coil_temp_sensor_fail
bit7	outdoor_temp_sensor_fail
bit8	condenser_temp_sensor_fail
bit9	indoor_temp_sensor_fail
bit10	discharge_temp_sensor_fail
bit11	/
bit12	indoor_fan_fault
bit13	outdoor_fan_fault
bit14	compressor_fault
bit15	electric_heater_fault
bit16	/
bit17	high_pressure_alarm
bit18	/
bit19	/
bit20	/
bit21	/
bit22	high_pressure_lockout
bit23	/
bit24	discharge_lockout
bit25	ac_over_voltage
bit26	ac_under_voltage
bit27	ac_power_loss
bit28	phase_loss
bit29	frequency_abnormal
bit30	phase_reverse
bit31	/

10.1.6. Liquid Cool Alarm

Bit Position	Register Name
bit0	High-temperature outlet water
bit1	Low-temperature outlet water
bit2	Water shortage alarm
bit3	Low water pressure differential
bit4	Low water level in the water tank
bit5	Excessive system pressure
bit6	Insufficient system pressure
bit7	Power Fault
bit8	Power Phase Voltage Fault
bit9	Power Voltage Fault
bit10	System Voltage High-Pressure Lockout
bit11	System Voltage Low-Pressure Lockout
bit12	Exhaust Over-temperature Lockout
bit13	Intake Air Temperature Sensor Malfunction
bit14	Abnormal Driving Voltage of the Compressor
bit15	Compressor Drive Lockout
bit16	Loss of Communication in the Compressor Driver
bit17	Abnormal Driving Voltage of the Water Pump
bit18	Water Pump Drive Lockout
bit19	Loss of Communication in the Water Pump Driver
bit20	Fault in the Outlet Water Temperature Sensor
bit21	Fault in the Inlet Water Temperature Sensor
bit22	Fault in the Suction Temperature Sensor
bit23~bit31	/

10.1.7. Fire-fighting Alarm

Bit Position	Register Name
bit0	Device Comm Fault
bit1	Device Failure
bit2	Product Expired
bit3	Device Over Voltage
bit4	Device Under Voltage
bit5	Fire-fighting Alarm
bit6	Water Leakage Alarm
bit7~31	/

10.2. PCS Alarm Code

10.2.1. Alarm HW

Bit Position	Register Name
bit0	AC module Alarm
bit1	Rack Func board Alarm
bit2	Overload alarm
bit3	Calibration parameter Abnormal
bit4~31	/

10.2.2. FaultAC

Bit Position	Register Name
bit0	EPO_Emergency Power Off
bit1	IGBT OCP Over Current Protection
bit2	Busbar Hardware Over Volt
bit3	Power Module cycle-by-cycle current limit
bit4	Balance Module cycle-by-cycle current limit
bit5	Fuse blown
bit6	AC module fault
bit7	Rack Func board fault
bit8	Main Board HW fault
bit9	Sampling zero Abnormal
bit10	Dry contact over-temperature
bit11	Auxiliary power Fault
bit12	Fan Fault
bit13	Connection Error
bit14	Inductor over-temperature
bit15	IGBT Module over temperature
bit16	Balance Module over temperature
bit17	System fire alarm
bit18	Ambient temperature over-temperature
bit19	Ambient humidity too high
bit20	Transformer over temp.
bit21	Hardware sampling Abnormal
bit22	Module current asymmetric
bit23~31	/

10.2.3. Fault AC

Bit Position	Register Name
bit0	Bus over Volt
bit1	Bus under Volt
bit2	Islanding protection
bit3	Bus Volt Abnormal
bit4	Bus over frequency
bit5	Bus under frequency
bit6	Phase reversed
bit7	Phase over current
bit8	Line N over current
bit9	Volt Imbalance
bit10	Current Imbalance
bit11	Phase lost
bit12	Access abnormal
bit13	Power down
bit14	Volt distortion
bit15	Pre-charge fault
bit16	Leakage current over current
bit17	Current DC component Abnormal
bit18	Off-grid output short-circuit
bit19	Off-grid Volt phase lost
bit20	Current harmonic Abnormal
bit21	Contactor fault
bit22	PE-N contactor short circuit
bit23	PE-N contactor open circuit
bit24~31	/

10.2.4. Fault DC

Bit Position	Register Name
bit0	Pre-charge fault
bit1	Input over Volt
bit2	Input under Volt
bit3	Input over current
bit4	Relay failure
bit5	DC input reversed
bit6	Positive and negative bus unbalanced
bit7	Balance module Over current
bit8	Low Battery
bit9	BMS Volt Abnormal
bit10	BMS current Abnormal
bit11	BMS temperature Abnormal
bit12~31	/

10.2.5. Fault SYS

Bit Position	Register Name
bit0	MainBoard Error
bit1	Insulation detection Abnormal
bit2	Ext input Dry-IO
bit3	Hardware version Fault
bit4	DSP initializing fault
bit5	SW and FW Mismatch
bit6	datalog data fault
bit7	Peripheral device failure
bit8	Parameter mismatch
bit9	Restart too much
bit10	SPD Surge Protective Device
bit11	Grounding fault
bit12	BMS shutdown failure
bit13	BMS battery status fault
bit14	BMS communication failure
bit15	PCS internal communication failure
bit16	EMS communication failure
bit17	STS communication failure
bit18	Parallel sys slave CAN communication fault
bit19	System resonance fault
bit20	High Volt ride-through timeout
bit21	Low Volt ride-through timeout
bit22	Synchronous signal fault
bit23	Switch_ON_OFF_Grid toggling error
bit24	Off-grid Volt phase reversed
bit25	PLL fault
bit26	/
bit27	/
bit28	/
bit29	/
bit30	/
bit31	Other fault